

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
DEPARTMENT OF MATHEMATICAL SCIENCES
SEMESTER I, SESSION 2026/2027

BMMS2094 STATISTICS FOR DATA SCIENCE
ASSIGNMENT (100%)

Instructions:

1. Students form groups of **3–4 members**.
2. Based on the concepts learned in class, prepare a report that applies one or more time series analysis techniques to a real-world dataset.
3. Each group is required to obtain its own dataset from a reliable source. The dataset should be recent and contain a sufficient number of observations for meaningful time series analysis.
4. Each group member must contribute **at least one time series forecasting model** to the analysis. Each individual report must clearly identify and discuss the time series model contributed by the respective student.
5. Students are expected to incorporate the techniques learned in class into their analysis and document their findings in the report. Meanwhile, students are encouraged to incorporate other relevant analytical techniques, such as descriptive statistics, linear regression, or other appropriate methods, to support their analysis.
6. The selected dataset and analysis should be related to **at least one of the United Nations Sustainable Development Goals (SDGs)**. Students should clearly identify the relevant SDG(s) and briefly explain how the analysis contributes to understanding or addressing the selected sustainability issue.
7. The report must be typed on a computer and comply with the following requirements.

Group Report

Each group shall prepare a paper containing the following:

- (a) Cover Page (indicate signatures and the percentage of contribution for each member)
- (b) Introduction (including background, objectives, dataset, SDG number and title)
- (c) Methodology (including dataset description, data preprocessing, overall analytical workflow, forecasting methods used, and forecast evaluation criteria, where applicable)
- (d) Data Analysis (combined results and discussion)
- (e) Conclusion (including discussion, limitations and recommendations, where appropriate)
- (f) Reference (IEEE referencing style)

Format: IEEE conference template.

Length: Maximum 5 pages.

Individual Report

Each student shall prepare an individual report containing the following:

- (a) Cover Page
- (b) Methodology (including the selected forecasting model, model specification/parameters, and justification for model selection)
- (c) Data Analysis (Results and Discussion)
- (d) Conclusion (including summary, limitations and recommendations, where appropriate)
- (e) Reference (APA referencing style)
- (f) Appendix (programming source code)

Format: Times New Roman, font size 12.

Length: Maximum 2 pages (excluding cover page, references, and appendix, if applicable)

Evaluation:

The assessment for this assignment consists of **group** and **individual** components.

Group Component

- Final Group Report

Individual Component

- Individual Report
- Presentation
- Peer Evaluation (each member will evaluate the contributions of his/her teammates)

The tutor will assess the group report, presentation, and individual report using the prescribed rubrics. Peer evaluation will be used to assess each student's contribution to the group work (refer to the *Peer Evaluation Form for Group Work*).

Submission Requirement

- All students are required to submit the following forms individually through Google Classroom:
 - (1) *AI Usage Disclosure Form*, and
 - (2) *Peer Evaluation Form for Group Work*.Students must not disclose their peer evaluation to other group members.
- The group leader is responsible for collecting the individual reports from all group members and submitting them together with the group report. Only one submission per group is required for the:
 - (1) individual reports, and
 - (2) group report.
- The group leader shall submit the files on behalf of the group. For the **group report**, complete the *Summary* sheet in the Excel file named “*BMMS2094 Assignment Assessment*” and attach the completed Excel file together with the report. Name the report and Excel file using the following format: *RDS2SIG3_Group5_GroupLeaderName.pdf* and *RDS2SIG3_Group5_GroupLeaderName.xlsx*.
- For the **individual reports**, the group leader must collect all individual reports in the final submission file. Name each individual report file using the following format: *RDS2SIG3_Group5_Name.pdf*

Late Submission

Late submission is NOT allowed. The following penalties apply to late submissions:

- Late 1 to 3 days after deadline of submission: minus 10 marks
- Late 4 to 7 days after deadline of submission: minus 20 marks
- Late more than 7 days after deadline of submission: 0 marks (assignment will not be accepted)

Peer Evaluation Form for Group Work

Your name : _____

Describe your role and job responsibilities in a group:

Write the name of each of your group members in a separate column. For each person, indicate the extent to which you agree with the criteria on the left, using a scale of 0-4 (0,1=unsatisfactory; 2=fair; 3=good; 4=outstanding) by referring to the *D. Peer Evaluation Rubrics*. Total the numbers in each column.

Evaluation Criteria	Group Member 1:	Group Member 2:	Group Member 3:
Working with team members			
Time Management			
Individual Contributions			
Professionalism and Attitude			
Leadership and Participation			
Total:			

Signature : _____

Date : _____

ASSIGNMENT ASSESSMENT RUBRICS**A. Group Report Rubric (40 Marks)**

Evaluation Criteria	Marks	Unsatisfactory (0–3)	Fair (4–5)	Good (6–7)	Outstanding (8)	Score
1. Introduction (Problem, Objectives, Dataset Source and SDG Alignment)	8	Problem statement is unclear or lacks objectives. Dataset source is unreliable, inappropriate, insufficient, or unsuitable for time series analysis. The relationship with SDG is unclear or absent.	Problem is identified but objectives lack clarity. Dataset source and suitability are briefly discussed but justification is limited. SDG relevance is mentioned but not well explained.	Problem and objectives are clearly stated. Dataset source is reliable, relevant, recent, and contains sufficient observations. The selected SDG is appropriately identified and explained.	Problem is clearly formulated with measurable objectives. Dataset is highly relevant, reliable, recent, and well justified. The selected SDG is strongly integrated with clear explanation of its significance.	
2. Methodology (Data Preparation, Forecasting Methods and Model Evaluation Approach)	8	Forecasting methods are inappropriate, incorrectly applied, or poorly explained. Little evidence of data preparation or methodological justification.	Appropriate methods are selected but explanations, preprocessing steps, or justification are incomplete. Model evaluation approach is unclear.	Appropriate forecasting methods are correctly implemented with suitable preprocessing, assumptions, and justification. Suitable evaluation measures are applied.	Methodology is comprehensive and technically sound. Data preparation, forecasting methods, model assumptions, diagnostics (where applicable), and evaluation approach are clearly justified and demonstrate excellent understanding.	
3. Data Analysis and Interpretation	8	Analysis is incomplete or contains major errors. Results are poorly presented, unsupported, or incorrectly interpreted.	Analysis is generally correct but interpretation lacks depth. Limited use of tables, graphs, or statistical evidence.	Analysis is accurate and supported by appropriate tables, visualisations, statistical evidence, and clear interpretation. Forecasting performance is evaluated using suitable measures.	Analysis is comprehensive and insightful. Forecasting models are critically analysed and compared where appropriate using suitable accuracy measures (e.g., MAE, RMSE, MAPE), statistical evidence, and effective visualisations.	
4. Conclusion and Critical Discussion (SDG Integration, Limitations and Recommendations)	8	Conclusion is unclear or missing. Findings are not critically discussed. SDG connection, limitations, and recommendations are absent.	Conclusion summarises findings with limited discussion. SDG linkage, limitations, or recommendations are briefly addressed.	Conclusion clearly discusses findings and relates them to the selected SDG. Limitations and practical recommendations are appropriately presented.	Conclusion demonstrates critical evaluation of findings, clearly integrates the SDG, acknowledges limitations, compares forecasting approaches where appropriate, and provides evidence-based recommendations with practical implications.	
5. Report Quality (IEEE Format, Organisation and Referencing)	8	Report lacks logical organisation. Frequent grammatical, formatting, citation, or IEEE referencing errors. Does not follow the IEEE paper template.	Report is generally organised but contains several language, formatting, or referencing errors. IEEE format is partially followed.	Report is well organised, clearly written, follows the IEEE paper template with only minor errors, and includes appropriate references.	Report is professionally written, logically structured, concise, visually effective, consistently follows the IEEE paper template, and contains accurate and complete IEEE references throughout.	
Total:						

B. Individual Report Rubric (20 Marks)

Evaluation Criteria	Marks	Unsatisfactory (0–1)	Fair (2–3)	Good (4)	Outstanding (5)	Score
1. Methodology (Model Selection, Parameters and Justification)	5	Methodology is unclear, incomplete, or inappropriate. The selected forecasting model, parameters, or implementation details are not adequately described.	Methodology is described but lacks sufficient explanation, justification, or discussion of model parameters.	Methodology is appropriate, correctly applied, and clearly explained. The selected forecasting model and parameters are adequately justified.	Methodology is comprehensive, technically sound, well justified, and demonstrates strong understanding of the selected forecasting technique, model parameters, and implementation.	
2. Data Analysis (Results and Discussion)	5	Forecasting analysis is incorrect, incomplete, or not implemented successfully. Results are poorly presented or interpreted.	Forecasting model is implemented with minor errors or limited interpretation of results.	Forecasting model is correctly implemented, interpreted, and evaluated using suitable measures. Results are clearly discussed.	Forecasting model is correctly implemented, critically interpreted, justified, and evaluated using suitable forecast accuracy measures. Results demonstrate strong analytical understanding.	
3. Conclusion (Summary, Limitations and Recommendations)	5	Conclusion is incomplete or missing. Summary of findings, limitations, and recommendations are not provided.	Conclusion is presented but discussion of findings, limitations, or recommendations is limited.	Conclusion appropriately summarises findings and includes relevant limitations and recommendations.	Conclusion critically discusses findings, clearly identifies limitations, and provides meaningful recommendations supported by the analysis.	
4. Report Quality (APA, Writing and Organisation)	5	Report lacks organisation. Frequent language, formatting, or APA referencing errors.	Report is organised but contains several APA or language errors.	Report is well organised, concise, and follows APA style with only minor errors.	Report is professionally written, concise, logically organised, and consistently follows APA formatting and referencing requirements.	
Total:						

C. Presentation Rubric (Individual Assessment) (20 Marks)

Evaluation Criteria	Marks	Unsatisfactory (0–1)	Fair (2)	Good (3)	Outstanding (4)	Score
1. Technical Understanding	4	Demonstrates limited understanding of the assigned forecasting model and project.	Demonstrates basic understanding but explanations lack clarity or depth.	Clearly explains the assigned methodology, analysis, and findings with good understanding.	Demonstrates comprehensive understanding of both the assigned forecasting model and the overall project, providing accurate and insightful explanations.	
2. Delivery and Communication	4	Speech is unclear, relies heavily on slides, and lacks audience	Delivery is understandable but hesitant or overly	Speaks clearly and confidently with effective audience	Delivers confidently and professionally, maintaining excellent communication and audience	

		engagement.	dependent on slides.	engagement.	engagement throughout.	
3. Explanation and Interpretation	4	Unable to explain results or interpret findings accurately.	Explains results with limited interpretation or justification.	Clearly explains and appropriately interprets the assigned analysis and results.	Critically interprets findings, justifies conclusions, and demonstrates strong analytical thinking.	
4. Responses to Questions	4	Unable to answer questions or provides inaccurate responses.	Answers basic questions with limited confidence or accuracy.	Responds accurately to most questions with good understanding.	Provides accurate, well-reasoned, and confident responses, demonstrating excellent mastery of the project.	
5. Contribution to Overall Group Presentation	4	Individual presentation disrupts the overall flow or shows little awareness of the group's presentation.	Contributes to the presentation but transitions or coordination with teammates are inconsistent.	Contributes effectively to a coherent group presentation with good coordination and smooth transitions.	Integrates seamlessly with the group presentation, demonstrates excellent coordination, supports teammates when appropriate, and enhances the overall professionalism of the presentation.	
Total:						

D. Peer Evaluation Rubric (20 Marks)

Evaluation Criteria	Marks	Unsatisfactory (0–1)	Fair (2)	Good (3)	Outstanding (4)
1. Working with Team Members	4	Rarely communicates or cooperates with team members.	Participates inconsistently in teamwork.	Communicates effectively and cooperates well with the team.	Consistently encourages collaboration, supports teammates, and contributes positively to group dynamics.
2. Time Management	4	Frequently misses deadlines or fails to complete assigned tasks.	Completes tasks but occasionally requires reminders.	Completes assigned tasks on time and manages workload effectively.	Completes tasks ahead of schedule, manages workload efficiently, and assists teammates when appropriate.
3. Individual Contribution	4	Makes little or no meaningful contribution to the project.	Completes assigned work with limited initiative.	Makes meaningful contributions, including completion of the assigned forecasting model and related tasks.	Makes significant contributions beyond assigned responsibilities and substantially improves the quality of the project.
4. Professionalism and Attitude	4	Displays poor attitude, lacks responsibility, or is uncooperative.	Generally cooperative but occasionally lacks professionalism or accountability.	Demonstrates a positive, respectful, and responsible attitude throughout the project.	Consistently demonstrates professionalism, accountability, integrity, and encourages positive team performance.
5. Leadership and Participation	4	Rarely participates or avoids responsibility.	Participates when requested but demonstrates limited initiative.	Actively participates, accepts responsibility, and contributes to decision-making.	Demonstrates initiative, coordinates team activities, solves problems proactively, and supports effective decision-making.